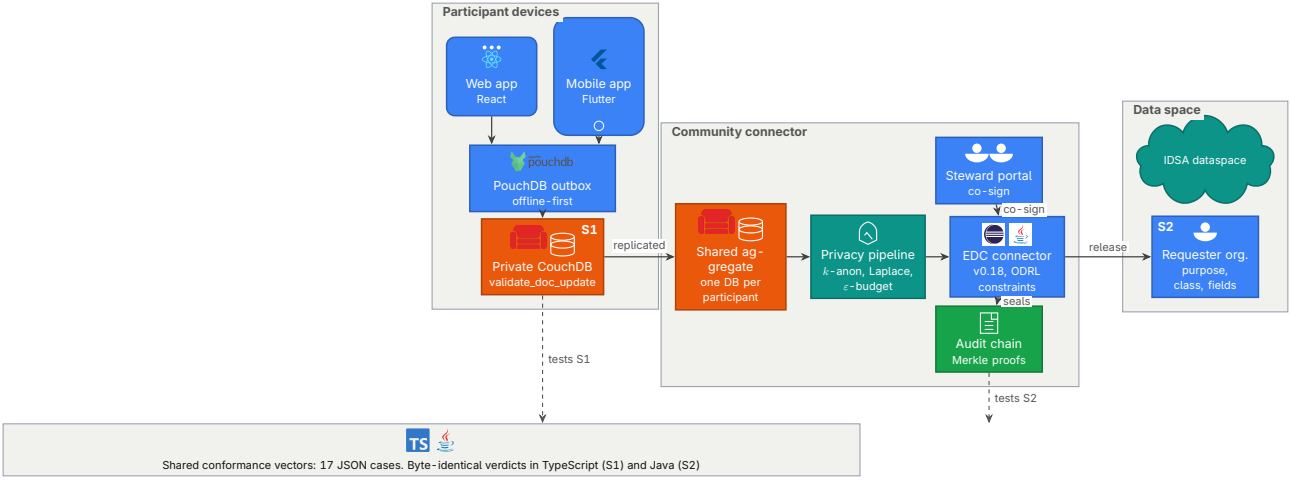
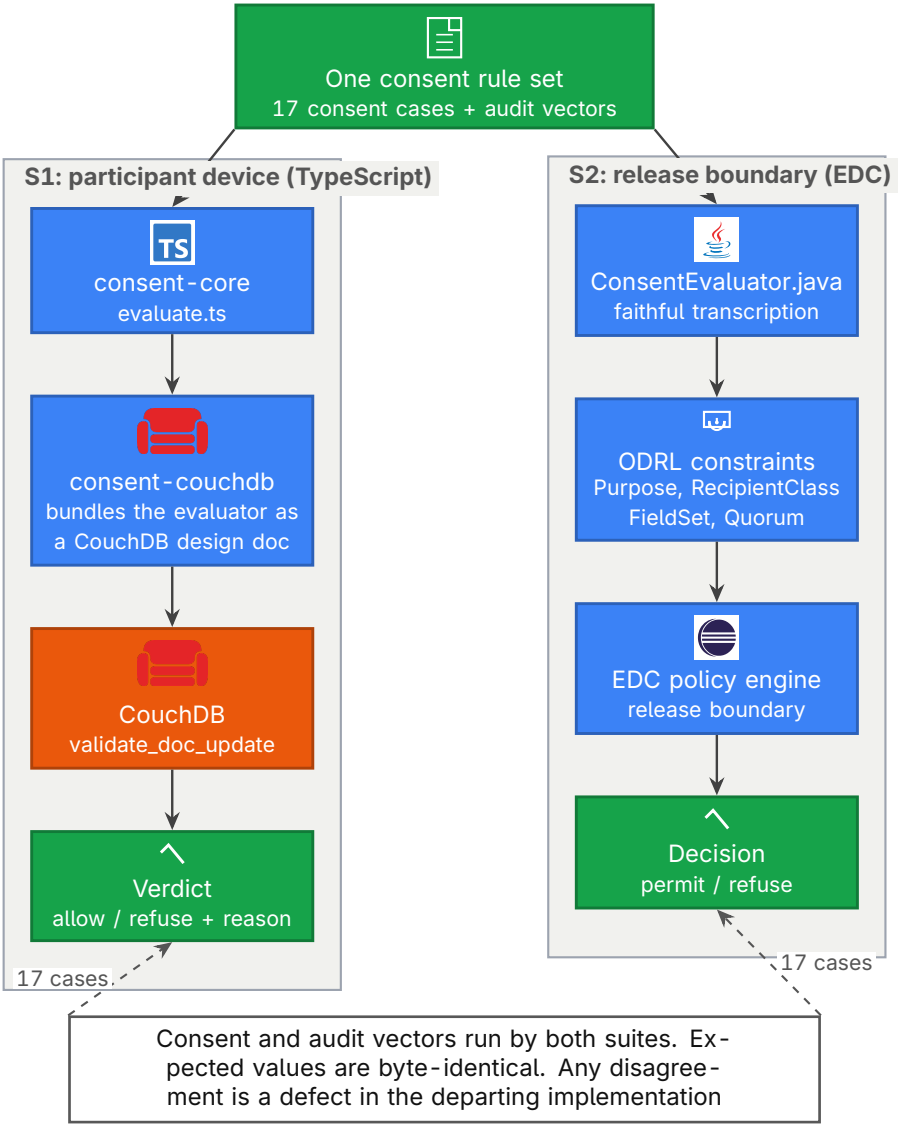
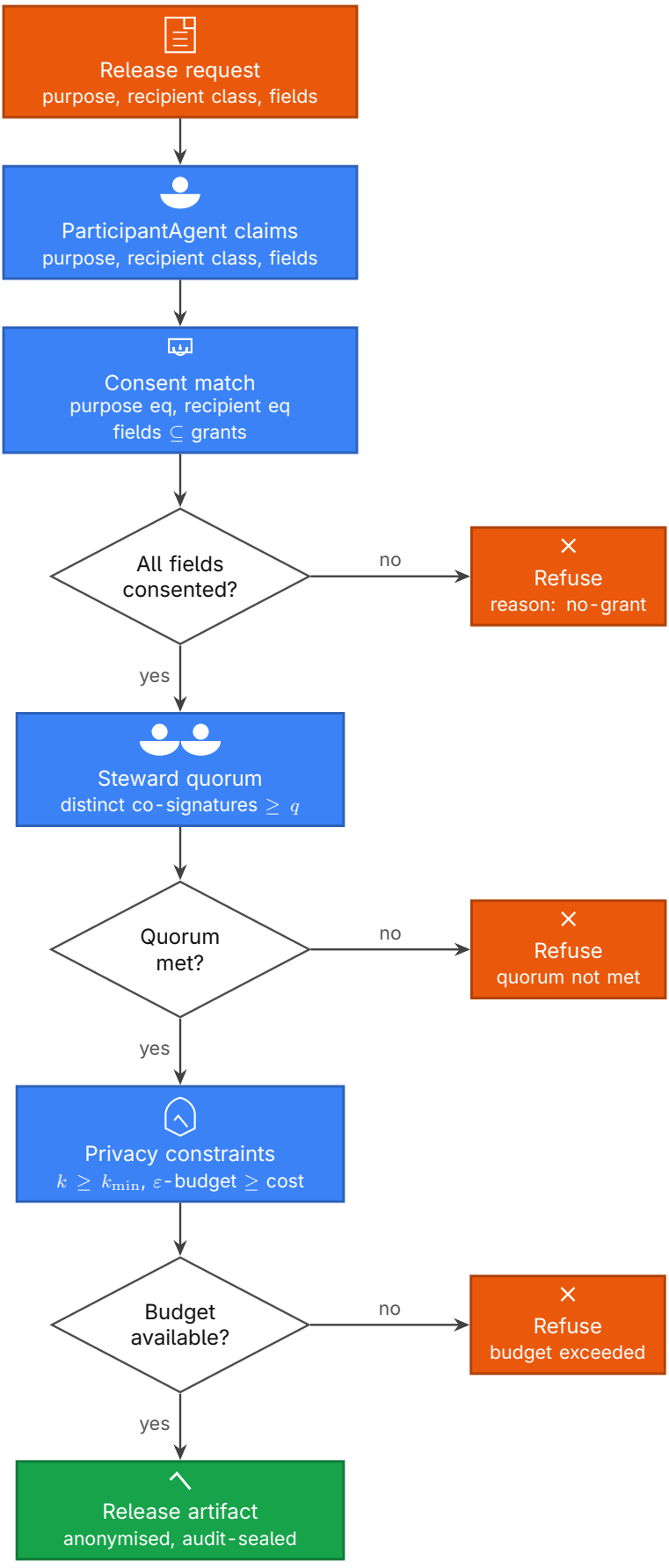
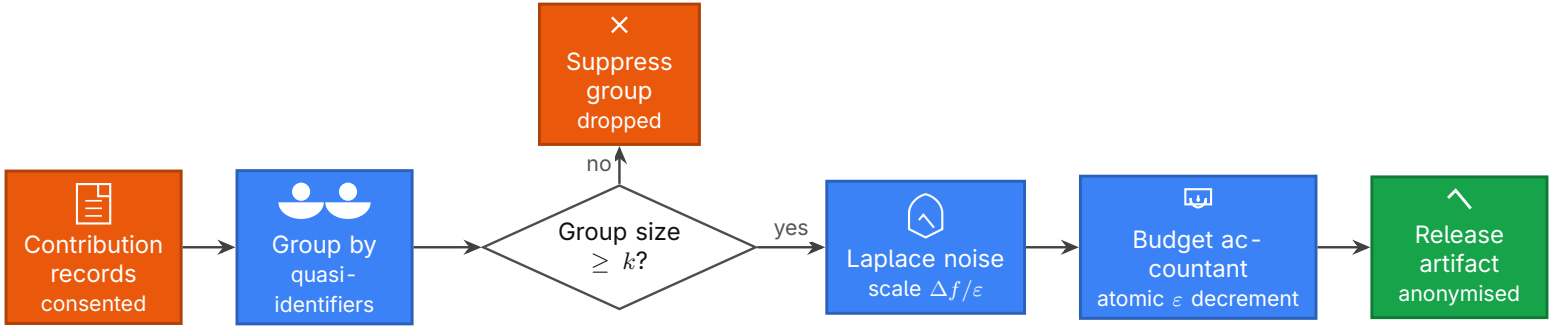

Participant devices
Web app
React
Mobile app
Flutter
pouchdb
PouchDB outbox
offline-first
S1
Private CouchDB
validate_doc_update
replicated
tests S1
Community connector
Shared aggregate
one DB per participant
Privacy pipeline
k-anon, Laplace, ε-budget
Steward portal
co-sign
co-sign
EDC connector
v0.18, ODRL constraints
seals
Audit chain
Merkle proofs
tests S2
release
Data space
IDSA dataspace
S2
Requester org.
purpose, class, fields
TS
Shared conformance vectors: 17 JSON cases. Byte-identical verdicts in TypeScript (S1) and Java (S2)


One consent rule set
17 consent cases + audit vectors
S1: participant device (TypeScript)
consent-core
evaluate.ts
consent-couchdb
bundles the evaluator as
a CouchDB design doc
CouchDB
validate_doc_update
Verdict
allow / refuse + reason
17 cases
S2: release boundary (EDC)
ConsentEvaluator.java
faithful transcription
ODRL constraints
Purpose, RecipientClass
FieldSet, Quorum
EDC policy engine
release boundary
Decision
permit / refuse
17 cases
Consent and audit vectors run by both suites. Expected values are byte-identical. Any disagreement is a defect in the departing implementation


Release request
purpose, recipient class, fields
ParticipantAgent claims
purpose, recipient class, fields
Consent match
purpose eq, recipient eq
fields ⊆ grants
All fields consented?
no
Refuse
reason: no-grant
yes
Steward quorum
distinct co-signatures ≥ q
Quorum met?
no
Refuse
quorum not met
yes
Privacy constraints
k ≥ k_min, ε-budget ≥ cost
Budget available?
no
Refuse
budget exceeded
yes
Release artifact
anonymised, audit-sealed


Contribution records consented
Group by quasi-identifiers
Group size $\geq k$?
no
Suppress group dropped
yes
Laplace noise scale $\Delta f/\varepsilon$
Budget ac-countant atomic ε decrement
Release artifact anonymised